RobustAI Engine — Executive Reliability Report

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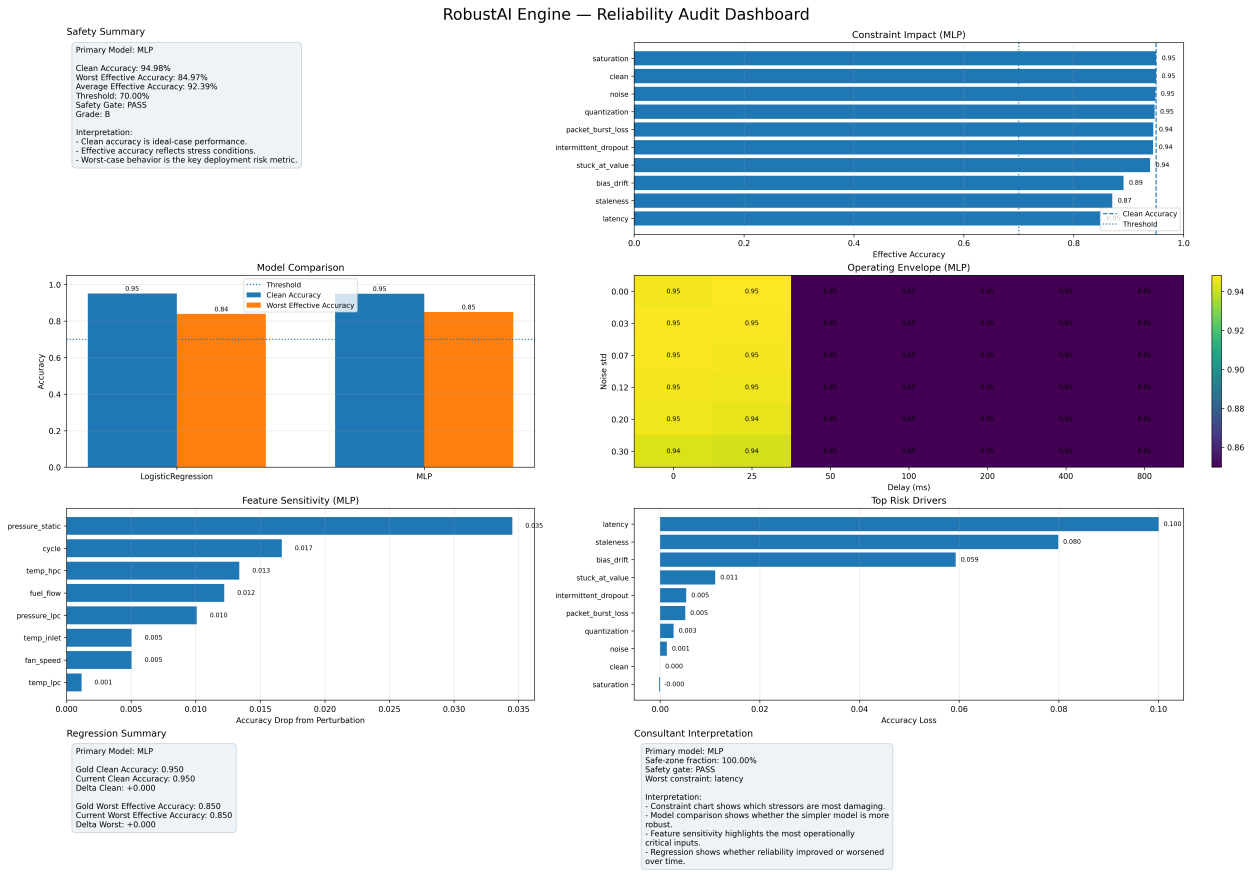
Operational stress audit with model comparison, safety gate, and mitigations

1) Reliability Dashboard



2) Audit Summary

RobustAI Engine Audit Report

Dataset: nasafd001
Profile: aviationextreme
Profile Description: High-altitude jet engine telemetry with tight latency deadlines and safety-critical monitoring.
Safety Threshold: 0.70
Best Clean Accuracy: 0.9506
Worst Effective Accuracy: 0.8389
Worst Case: model=LogisticRegression, constraint=staleness
PASS

Safety gate did not trigger for this run.
Result Summary

model	profile	constraint	cleanaccuracy	effectiveaccuracy	delta
LogisticRegression	aviationextreme	clean	0.950562	0.950562	0
LogisticRegression	aviationextreme	noise	0.950562	0.950756	-0.000193874
LogisticRegression	aviationextreme	noise	0.950562	0.949787	0.000775494
LogisticRegression	aviationextreme	noise	0.950562	0.949399	0.00116324
LogisticRegression	aviationextreme	noise	0.950562	0.948236	0.00232648
LogisticRegression	aviationextreme	latency	0.950562	0.945909	0.00465297
LogisticRegression	aviationextreme	staleness	0.950562	0.94843	0.00213261
LogisticRegression	aviationextreme	biasdrift	0.950562	0.949593	0.000969368
LogisticRegression	aviationextreme	latency	0.950562	0.849748	0.100814
LogisticRegression	aviationextreme	staleness	0.950562	0.94746	0.00310198
LogisticRegression	aviationextreme	biasdrift	0.950562	0.947073	0.00348972
LogisticRegression	aviationextreme	latency	0.950562	0.849748	0.100814
LogisticRegression	aviationextreme	staleness	0.950562	0.940093	0.0104692
LogisticRegression	aviationextreme	biasdrift	0.950562	0.944746	0.00581621
LogisticRegression	aviationextreme	latency	0.950562	0.849748	0.100814
LogisticRegression	aviationextreme	staleness	0.950562	0.912369	0.0381931
LogisticRegression	aviationextreme	biasdrift	0.950562	0.934858	0.0157038
LogisticRegression	aviationextreme	latency	0.950562	0.849748	0.100814
LogisticRegression	aviationextreme	staleness	0.950562	0.838891	0.111671
LogisticRegression	aviationextreme	biasdrift	0.950562	0.900543	0.0500194
LogisticRegression	aviationextreme	intermittentdropout	0.950562	0.944746	0.00581621
LogisticRegression	aviationextreme	stuckatvalue	0.950562	0.939705	0.0108569
LogisticRegression	aviationextreme	saturation	0.950562	0.951144	-0.000581621
LogisticRegression	aviationextreme	quantization	0.950562	0.94746	0.00310198
LogisticRegression	aviationextreme	packetburstloss	0.950562	0.945522	0.00504071
MLP	aviationextreme	clean	0.949787	0.949787	0
MLP	aviationextreme	noise	0.949787	0.949205	0.000581621
MLP	aviationextreme	noise	0.949787	0.948817	0.000969368
MLP	aviationextreme	noise	0.949787	0.94843	0.00135712
MLP	aviationextreme	noise	0.949787	0.949399	0.000387747
MLP	aviationextreme	latency	0.949787	0.94494	0.00484684
MLP	aviationextreme	staleness	0.949787	0.950174	-0.000387747
MLP	aviationextreme	biasdrift	0.949787	0.947848	0.00193874
MLP	aviationextreme	latency	0.949787	0.849748	0.100039
MLP	aviationextreme	staleness	0.949787	0.949399	0.000387747
MLP	aviationextreme	biasdrift	0.949787	0.946103	0.0036836

MLP	aviationextreme	latency	0.949787	0.849748	0.100039
MLP	aviationextreme	staleness	0.949787	0.94145	0.00833656
MLP	aviationextreme	biasdrift	0.949787	0.941256	0.00853044
MLP	aviationextreme	latency	0.949787	0.849748	0.100039
MLP	aviationextreme	staleness	0.949787	0.913726	0.0360605
MLP	aviationextreme	biasdrift	0.949787	0.931369	0.018418
MLP	aviationextreme	latency	0.949787	0.849748	0.100039
MLP	aviationextreme	staleness	0.949787	0.869911	0.0798759
MLP	aviationextreme	biasdrift	0.949787	0.890461	0.0593253
MLP	aviationextreme	intermittentdropout	0.949787	0.944552	0.00523459
MLP	aviationextreme	stuckatvalue	0.949787	0.938736	0.0110508
MLP	aviationextreme	saturation	0.949787	0.949981	-0.000193874
MLP	aviationextreme	quantization	0.949787	0.947073	0.00271423
MLP	aviationextreme	packetburstloss	0.949787	0.944746	0.00504071

3) Regression Summary

Regression Summary

model	constraint	effectiveaccuracygold	effectiveaccuracycurrent	deltavsgold
LogisticRegression	biasdrift	0.950756	0.949593	-0.00116324
LogisticRegression	biasdrift	0.950756	0.947073	-0.0036836
LogisticRegression	biasdrift	0.950756	0.944746	-0.00601008
LogisticRegression	biasdrift	0.950756	0.934858	-0.0158976
LogisticRegression	biasdrift	0.950756	0.900543	-0.0502133
LogisticRegression	biasdrift	0.950368	0.949593	-0.000775494
LogisticRegression	biasdrift	0.950368	0.947073	-0.00329585
LogisticRegression	biasdrift	0.950368	0.944746	-0.00562233
LogisticRegression	biasdrift	0.950368	0.934858	-0.0155099
LogisticRegression	biasdrift	0.950368	0.900543	-0.0498255
LogisticRegression	biasdrift	0.949593	0.949593	0
LogisticRegression	biasdrift	0.949593	0.947073	-0.00252036
LogisticRegression	biasdrift	0.949593	0.944746	-0.00484684
LogisticRegression	biasdrift	0.949593	0.934858	-0.0147344
LogisticRegression	biasdrift	0.949593	0.900543	-0.04905
LogisticRegression	biasdrift	0.947073	0.949593	0.00252036
LogisticRegression	biasdrift	0.947073	0.947073	0
LogisticRegression	biasdrift	0.947073	0.944746	-0.00232648
LogisticRegression	biasdrift	0.947073	0.934858	-0.012214
LogisticRegression	biasdrift	0.947073	0.900543	-0.0465297
LogisticRegression	clean	0.950562	0.950562	0
LogisticRegression	intermittentdropout	0.949787	0.944746	-0.00504071
LogisticRegression	latency	0.948817	0.945909	-0.0029081
LogisticRegression	latency	0.948817	0.849748	-0.0990694
LogisticRegression	latency	0.948817	0.849748	-0.0990694
LogisticRegression	latency	0.948817	0.849748	-0.0990694
LogisticRegression	latency	0.948817	0.849748	-0.0990694
LogisticRegression	latency	0.904808	0.945909	0.0411012
LogisticRegression	latency	0.904808	0.849748	-0.0550601
LogisticRegression	latency	0.904808	0.849748	-0.0550601
LogisticRegression	latency	0.904808	0.849748	-0.0550601
LogisticRegression	latency	0.904808	0.849748	-0.0550601
LogisticRegression	latency	0.849748	0.945909	0.0961613
LogisticRegression	latency	0.849748	0.849748	0
LogisticRegression	latency	0.849748	0.849748	0
LogisticRegression	latency	0.849748	0.849748	0
LogisticRegression	latency	0.849748	0.849748	0
LogisticRegression	latency	0.849748	0.945909	0.0961613
LogisticRegression	latency	0.849748	0.849748	0
LogisticRegression	latency	0.849748	0.849748	0
LogisticRegression	latency	0.849748	0.849748	0
LogisticRegression	latency	0.849748	0.849748	0
LogisticRegression	noise	0.950756	0.950756	0
LogisticRegression	noise	0.950756	0.949787	-0.000969368
LogisticRegression	noise	0.950756	0.949399	-0.00135712
LogisticRegression	noise	0.950756	0.948236	-0.00252036
LogisticRegression	noise	0.949981	0.950756	0.000775494
LogisticRegression	noise	0.949981	0.949787	-0.000193874
LogisticRegression	noise	0.949981	0.949399	-0.000581621

LogisticRegression	noise	0.949981	0.948236	-0.00174486
LogisticRegression	noise	0.949011	0.950756	0.00174486
LogisticRegression	noise	0.949011	0.949787	0.000775494
LogisticRegression	noise	0.949011	0.949399	0.000387747
LogisticRegression	noise	0.949011	0.948236	-0.000775494
LogisticRegression	packetburstloss	0.948623	0.945522	-0.00310198
LogisticRegression	quantization	0.95095	0.94746	-0.00348972
LogisticRegression	saturation	0.950756	0.951144	0.000387747
LogisticRegression	staleness	0.950756	0.94843	-0.00232648
LogisticRegression	staleness	0.950756	0.94746	-0.00329585
LogisticRegression	staleness	0.950756	0.940093	-0.010663
LogisticRegression	staleness	0.950756	0.912369	-0.038387
LogisticRegression	staleness	0.950756	0.838891	-0.111865
LogisticRegression	staleness	0.949787	0.94843	-0.00135712
LogisticRegression	staleness	0.949787	0.94746	-0.00232648
LogisticRegression	staleness	0.949787	0.940093	-0.00969368
LogisticRegression	staleness	0.949787	0.912369	-0.0374176
LogisticRegression	staleness	0.949787	0.838891	-0.110896
LogisticRegression	staleness	0.948236	0.94843	0.000193874
LogisticRegression	staleness	0.948236	0.94746	-0.000775494
LogisticRegression	staleness	0.948236	0.940093	-0.00814269
LogisticRegression	staleness	0.948236	0.912369	-0.0358666
LogisticRegression	staleness	0.948236	0.838891	-0.109345
LogisticRegression	staleness	0.946297	0.94843	0.00213261
LogisticRegression	staleness	0.946297	0.94746	0.00116324
LogisticRegression	staleness	0.946297	0.940093	-0.00620396
LogisticRegression	staleness	0.946297	0.912369	-0.0339279
LogisticRegression	staleness	0.946297	0.838891	-0.107406
LogisticRegression	stuckatvalue	0.94843	0.939705	-0.00872431
MLP	biasdrift	0.949205	0.947848	-0.00135712
MLP	biasdrift	0.949205	0.946103	-0.00310198
MLP	biasdrift	0.949205	0.941256	-0.00794882
MLP	biasdrift	0.949205	0.931369	-0.0178364
MLP	biasdrift	0.949205	0.890461	-0.0587437
MLP	biasdrift	0.949205	0.947848	-0.00135712
MLP	biasdrift	0.949205	0.946103	-0.00310198
MLP	biasdrift	0.949205	0.941256	-0.00794882
MLP	biasdrift	0.949205	0.931369	-0.0178364
MLP	biasdrift	0.949205	0.890461	-0.0587437
MLP	biasdrift	0.947848	0.947848	0
MLP	biasdrift	0.947848	0.946103	-0.00174486
MLP	biasdrift	0.947848	0.941256	-0.0065917
MLP	biasdrift	0.947848	0.931369	-0.0164793
MLP	biasdrift	0.947848	0.890461	-0.0573866
MLP	biasdrift	0.946103	0.947848	0.00174486
MLP	biasdrift	0.946103	0.946103	0
MLP	biasdrift	0.946103	0.941256	-0.00484684
MLP	biasdrift	0.946103	0.931369	-0.0147344
MLP	biasdrift	0.946103	0.890461	-0.0556417
MLP	clean	0.949787	0.949787	0
MLP	intermittentdropout	0.949399	0.944552	-0.00484684
MLP	latency	0.947848	0.94494	-0.0029081
MLP	latency	0.947848	0.849748	-0.0981
MLP	latency	0.947848	0.849748	-0.0981
MLP	latency	0.947848	0.849748	-0.0981

MLP	latency	0.947848	0.849748	-0.0981
MLP	latency	0.903839	0.94494	0.0411012
MLP	latency	0.903839	0.849748	-0.0540907
MLP	latency	0.903839	0.849748	-0.0540907
MLP	latency	0.903839	0.849748	-0.0540907
MLP	latency	0.903839	0.849748	-0.0540907
MLP	latency	0.849748	0.94494	0.0951919
MLP	latency	0.849748	0.849748	0
MLP	latency	0.849748	0.849748	0
MLP	latency	0.849748	0.849748	0
MLP	latency	0.849748	0.849748	0
MLP	latency	0.849748	0.94494	0.0951919
MLP	latency	0.849748	0.849748	0
MLP	latency	0.849748	0.849748	0
MLP	latency	0.849748	0.849748	0
MLP	latency	0.849748	0.849748	0
MLP	noise	0.950174	0.949205	-0.000969368
MLP	noise	0.950174	0.948817	-0.00135712
MLP	noise	0.950174	0.94843	-0.00174486
MLP	noise	0.950174	0.949399	-0.000775494
MLP	noise	0.948817	0.949205	0.000387747
MLP	noise	0.948817	0.948817	0
MLP	noise	0.948817	0.94843	-0.000387747
MLP	noise	0.948817	0.949399	0.000581621
MLP	noise	0.949205	0.949205	0
MLP	noise	0.949205	0.948817	-0.000387747
MLP	noise	0.949205	0.94843	-0.000775494
MLP	noise	0.949205	0.949399	0.000193874
MLP	packetburstloss	0.947848	0.944746	-0.00310198
MLP	quantization	0.949787	0.947073	-0.00271423
MLP	saturation	0.949787	0.949981	0.000193874
MLP	staleness	0.949787	0.950174	0.000387747
MLP	staleness	0.949787	0.949399	-0.000387747
MLP	staleness	0.949787	0.94145	-0.00833656
MLP	staleness	0.949787	0.913726	-0.0360605
MLP	staleness	0.949787	0.869911	-0.0798759
MLP	staleness	0.950174	0.950174	0
MLP	staleness	0.950174	0.949399	-0.000775494
MLP	staleness	0.950174	0.94145	-0.00872431
MLP	staleness	0.950174	0.913726	-0.0364482
MLP	staleness	0.950174	0.869911	-0.0802637
MLP	staleness	0.949593	0.950174	0.000581621
MLP	staleness	0.949593	0.949399	-0.000193874
MLP	staleness	0.949593	0.94145	-0.00814269
MLP	staleness	0.949593	0.913726	-0.0358666
MLP	staleness	0.949593	0.869911	-0.079682
MLP	staleness	0.946491	0.950174	0.0036836
MLP	staleness	0.946491	0.949399	0.0029081
MLP	staleness	0.946491	0.94145	-0.00504071
MLP	staleness	0.946491	0.913726	-0.0327646
MLP	staleness	0.946491	0.869911	-0.0765801
MLP	stuckatvalue	0.947654	0.938736	-0.00891819

4) Mitigation Recommendations

Mitigation Recommendations

• Latency is a major failure mode. Consider buffering, timeout-aware inference, or fallback logic.

• Staleness is hurting reliability. Consider freshness checks, timestamp validation, or drift-aware preprocessing.